Problem-Solving Insights II

Small Value Approximations

Lexington Physics Club

A Visit to the Tiny Workshop

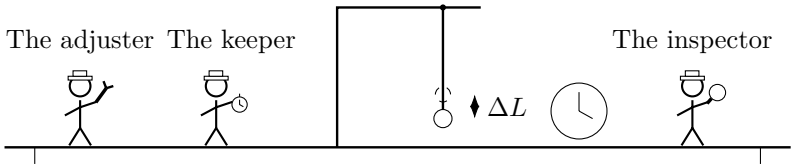
In everyday life, we calibrate by adjusting things little by little. However, we can't adjust results manually (which is rather unfortunate). If we treat life as a function, all we can do is to change our input variables, and often very slightly.

Then we ask ourselves: "How much difference will it make in the end? Is it worth it?"

Physics has its own version of this question. For example when a measured time is slightly off, or a ruler is too coarse. We want to know how these small differences—which could sometimes be uncertain—travel through an experiment or expression.

Imagine that every experiment has a tiny workshop hidden inside it. One tinkerer carries a wrench and adjusts the apparatus; another carries a ruler and a stopwatch; a third inspects the calculation.

Once upon a time, their project was a pendulum clock. The adjuster lengthened its string a tiny bit. The keeper watched the clock as usual. The inspector asked the keeper: "Hey do you see anything strange today?"



The length change is exaggerated so that we can see it.

At first, this sounds like nothing more than rounding numbers. Yet a mathematically bound approximation can tell us somewhat more logically about how things will change, avoiding heuristic guessing and awkward calculations at the same time.

In the Tiny Workshop

For each puzzle, make a prediction from intuitions before doing any math.

1. Small compared with what?

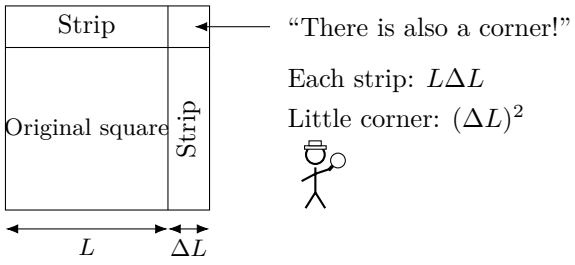
The adjuster adds 1 mm to a 1 m pendulum string. In a different apparatus, the adjuster widens a 2 mm candy design by the same 1 mm. Which change is more outstanding to you?

Mathematically, how would you describe the extent of a change?

To see why the comparison matters, let's draw out a small change.

2. A slightly larger square

A square sheet has side length L. The adjuster increases each side by 1%. Does its area increase by approximately 1%, 2%, or 4%? Explain your prediction using the extra pieces in the picture.



**The added width is exaggerated.*

Some math you may use:

Set the relative change as $\varepsilon = \Delta L/L$. Therefore, the new area is

$$(L + \Delta L)^2 = L^2(1 + 2\varepsilon + \varepsilon^2).$$

For $|\varepsilon| \ll 1$ ($\ll$ means "much smaller"), the corner's contribution, ε^2, is much smaller than the strips' contribution, 2ε. Keeping the constant and linear terms is called a *first-order approximation.*

The corner is of course still there. We have only decided that it is small enough to leave out of this estimate. But sometimes, approximation to first order wouldn't work...

3. Undoing a change—almost

A string is lengthened by 5%, then shortened by 5% of its *new* length. Is it back to its original length? Examine $L_{\text{final}} = L(1 + \varepsilon)(1 - \varepsilon)$.

A step further. Since we stressed on **first** order approximation previously, there must be other orders as well. What do you think is the definition for "order"? Using that definition, What order did you keep in this problem?

4. LHS Bells pmo...

Back in the workshop, the adjuster shortens the pendulum string by 2%, while g (gravity) stays fixed. For small swings, the period is

$$T = 2\pi\sqrt{\frac{L}{g}}.$$

Does the period decrease by more than 2%, less than 2%, or exactly 2%? Start with a prediction, then use

$$\frac{T_{\text{new}}}{T_{\text{old}}} = \sqrt{1+\varepsilon}.$$

Here is a useful rule: for a fixed exponent n and sufficiently small $|\varepsilon|$,

$$(1+\varepsilon)^n \approx 1 + n\varepsilon.$$

In particular, $\sqrt{1+\varepsilon} \approx 1 + \varepsilon/2$.

A step further. A clock was calibrated using the old period. Its mechanism still counts the same number of swings as "one minute." After the string is shortened, does the clock run fast or slow? Approximately how much time does it lose or gain in one actual hour?

What if the tinkerer gets mad today and changes two things at once?

5. Two changes, one result

Same setup as in the last problem, but this time L increases by 3% and g increases by 1%.

Do these changes push the period in the same direction or in opposite directions? Estimate their combined effect using

$$\frac{T_{\text{new}}}{T_{\text{old}}} = (1+0.03)^{1/2}(1+0.01)^{-1/2}.$$

(Hint: Multiply your first-order approximations, then leave out the product of the two small relative changes, just like how to dealt with the square)

A step further. Suppose L and g instead increase by exactly the same percentage. Does the period remain unchanged exactly?

The keeper can predict a change when the adjuster tells us exactly what happened. But a measurement does not always tell us an exact input.

6. How well do we know the predicted period?

The keeper measures a length of 1.000 m and estimates that the true length lies between 0.990 m and 1.010 m. Take g as known exactly and fixed.

Approximately what percentage range does this give for the predicted period? Use the period at 1.000 m as your reference.

A step further. Now we measure T and calculate $g = 4\pi^2 L/T^2$. If our measured period is 1% too large and L is correct, is our calculated g too large or too small? By approximately what percentage?

A small adjustment changes the apparatus. An uncertain measurement limits how precisely we know it. Although they mean different things, mathematically we describe them in a similar way.

7. Lagrangian Mechanics at Play?

A pendulum of length L moves through a small angle θ, measured in radians. Its rise in height is

$$h = L(1 - \cos\theta).$$

The adjuster uses $\cos\theta \approx 1$ and concludes that $h \approx 0$. The inspector objects. What went wrong?

Use the approximation $\cos\theta \approx 1 - \theta^2/2$ to find the height change to the smallest order.

A step further. If the angle doubles while remaining small, approximately what happens to the rise in height? How does that compare with the horizontal displacement, $x = L\sin\theta \approx L\theta$?

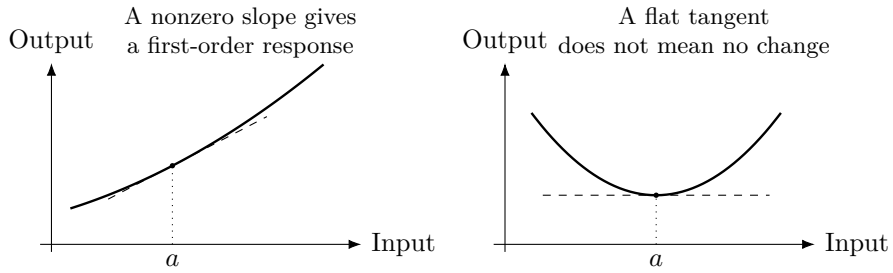
The Small Detail Can Be the Whole Physics

Our tinkerers have learned to follow a small change through an experiment. Sometimes a first-order estimate is enough. Sometimes two changes cancel, or a tiny difference accumulates. Sometimes the first-order term vanishes, and the detail we cannot ignore is the "largest small term" that tells us the change.

Now you will see the mathematical framework behind these approximations.

Let first consider a simple case: Near a chosen point, a smooth curve often looks almost like a line. That line's slope tells us how strongly the output responds to a small change in the input.

But if the line is flat, we must look more closely.



One Knob v.s. Several Knobs

If you know derivatives, the straight-line approximation is

$$f(a + \delta) \approx f(a) + f'(a)\delta.$$

Here, δ is a small change in the input, and $f'(a)$ is the slope at our starting point. If that slope is zero, the first-order term tells us nothing about the change. We may need the next term:

$$f(a + \delta) \approx f(a) + f'(a)\delta + \frac{1}{2}f''(a)\delta^2.$$

If that second order term is also zero, we then need the next term... Until we stop at the first non-zero order term.

What if the adjuster turns several knobs and change many values together? Suppose an output depends on n inputs:

$$F = F(x_1, x_2, \ldots, x_n).$$

Let the starting inputs be $\mathbf{a} = (a_1, \ldots, a_n)$, and let their small changes be $\boldsymbol{\delta} = (\delta_1, \ldots, \delta_n)$. The general expansion through second order is

$$F(\mathbf{a} + \boldsymbol{\delta}) \approx F(\mathbf{a}) + \sum_{i=1}^{n} \left.\frac{\partial F}{\partial x_i}\right|_{\mathbf{a}} \delta_i + \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \left.\frac{\partial^2 F}{\partial x_i \, \partial x_j}\right|_{\mathbf{a}} \delta_i \delta_j.$$

Despite the busy notation (you would never need this in physics olympiads anyway), the first-order rule still holds: find the effect of each small input change while keeping the other inputs fixed, then add those effects.

In other words, each partial derivative is the slope associated with one variable.

At second order, we also keep squares of small changes and products of different small changes. For two inputs, the second-order contribution is

$$\frac{1}{2} F_{xx} \delta_x^2 + F_{xy} \delta_x \delta_y + \frac{1}{2} F_{yy} \delta_y^2.$$

The middle term considers how the effect of changing one input may depend on the other input.

For our pendulum, the first-order rule becomes

$$\frac{\Delta T}{T} \approx \frac{1}{2} \frac{\Delta L}{L} - \frac{1}{2} \frac{\Delta g}{g}.$$

A longer string increases the period; stronger gravity decreases it.

A Tiny Change in Time

Imagine that the keeper takes a clock onto a train. The inspector sits still on a bench at the train station.

Special relativity from our last lecture, tells us that, in an inertial frame where the clock moves at speed v, it records less elapsed time than what the inspector would see on a clock at the station.

Let t be the elapsed time in the station and τ be the elapsed time recorded by the clock on the train. Then

$$t = \gamma \tau, \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}},$$

where $c \approx 3.00 \times 10^8 \, \mathrm{m/s}$ is the speed of light.

If we write $\beta = v/c$, our familiar approximation returns:

$$\gamma = (1 - \beta^2)^{-1/2} \approx 1 + \frac{1}{2} \beta^2, \qquad |\beta| \ll 1.$$

How much does this affect an ordinary journey? Take an illustrative speed of 30 m/s, or 108 km/h. Then

$$\frac{v}{c} \approx 10^{-7}, \qquad \frac{1}{2}\left(\frac{v}{c}\right)^2 \approx 5 \times 10^{-15}.$$

Since

$$\tau = t\sqrt{1 - \frac{v^2}{c^2}} \approx t\left(1 - \frac{v^2}{2c^2}\right),$$

the elapsed-time difference is approximately

$$t - \tau \approx t\,\frac{v^2}{2c^2}.$$

For a clock moving at this constant speed for a full 24 hours,

$$t - \tau \approx (86400\,\text{s})(5 \times 10^{-15}) \approx 4.3 \times 10^{-10}\,\text{s}.$$

That is about 0.43 nanoseconds—less than one billionth of a second.

This is why relativity is almost impossible to be realized from daily experiences.

An Inspector Calls[1]

"Good evening. I understand that someone here has been neglecting small terms.

"You see, I am not here to insist that every detail be kept. That would make me a rather tiresome guest. I am here to ask whether you know what you have discarded, and whether your answer can survive its absence.

"Before you take your next shortcut, I have four questions. Just a routine inspection you should take note of."

- "You say it is small. Small compared with what?"
- "What, precisely, and mathematically, are you trying to find?"
- "Have you considered all the sources of change?"
- "Which is the first nonzero term that tells you about change?"

"That will be all. You may return to your calculations."

Somewhere in the workshop, a tinkerer quietly scribbles down a second-order term.

[1] A nod to J. B. Priestley's play *An Inspector Calls*, first performed in 1945, in which a mysterious inspector questions a family about the consequences of their actions. Our inspector is concerned with the consequences of neglected terms instead.